Chapter 1: Quantum Mechanics

The wave-particle duality is a fundamental concept in quantum mechanics. It states that every particle or qu

The Schrodinger equation is the fundamental equation of quantum mechanics that describes how the quant

The Heisenberg Uncertainty Principle states that it is impossible to simultaneously measure both the positio

Chapter 2: Semiconductor Physics

Band theory describes the behavior of electrons in solids. In a crystalline solid, the allowed energy levels of

The valence band is the highest range of electron energies in which electrons are normally present at absolu

A PN junction is formed when a p-type semiconductor is joined with an n-type semiconductor. The p-type re